

Prediction context

Anchor history S_0
and query x

Candidate pool
 $C = \{C_1, \dots, C_K\}$

Fixed predictor f
(trained once, then frozen)

Response-aware marginal router

Cached screening
 $\hat{m}_C(j | A)$

Predictor response on shortlist
 $f_A, f_{A \cup \{j\}} \rightarrow r_{A,j}$

Response-aware marginal scorer
 $\hat{m}_R(j | A)$

Outcome-supervised training

Observed marginal
 $\ell(f_A(x), y) - \ell(f_{A \cup \{j\}}(x), y)$

Regression + gap-weighted
ranking objective

Sequential selection
select, then stop when $\hat{m}_R \leq 0$

deploy: update $A \leftarrow A \cup \{j^*\}$ and repeat